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Assignment 1: Linux Basic Commands

Section A: Navigation

1. mkdir linux_practice
2. cd linux_practice
3. mkdir test_folder
4. cd ..
5. Pwd
6. ls, ls -l
7. man ls , ls --help

```
root@gourshab:/home# mkdir linux_practice
root@gourshab:/home# cd linux_practice
root@gourshab:/home/linux_practice# mkdir test_folder
root@gourshab:/home/linux_practice# ls
test_folder
root@gourshab:/home/linux_practice# cd ..
root@gourshab:/home# pwd
/home
root@gourshab:/home# ls
linux_practice
root@gourshab:/home# ls -l
total 4
drwxr-xr-x 3 root root 4096 Mar 15 15:05 linux_practice
root@gourshab:/home# man ls
root@gourshab:/home# ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                               e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                               modification of file status information);
                               with -l: show ctime and sort by name;
                               otherwise: sort by ctime, newest first
  -C                         list entries by columns
      --color[=WHEN]         colorize the output; WHEN can be 'always' (default
                               if omitted), 'auto', or 'never'; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         do not sort, enable -aU, disable -ls --color
  -F, --classify             append indicator (one of */=>@|) to entries
      --file-type            likewise, except do not append '*'
      --format=WORD          across -x, commas -m, horizontal -x, long -l,
                               single-column -1, verbose -l, vertical -C
```

Section B: File & Directory Operations

1. touch student.txt

2. mkdir data
3. cp student.txt data/
4. mv student.txt info.txt
5. rm info.txt
6. rmdir data , rm -rf data

```
root@gourshab:/home# ls
linux_practice
root@gourshab:/home# touch student.txt
root@gourshab:/home# ls
linux_practice  student.txt
root@gourshab:/home# mkdir data
root@gourshab:/home# ls
data  linux_practice  student.txt
root@gourshab:/home# cp student.txt data/
root@gourshab:/home# ls
data  linux_practice  student.txt
root@gourshab:/home# mv student.txt info.txt
root@gourshab:/home# ls
data  info.txt  linux_practice
root@gourshab:/home# rm info.txt
root@gourshab:/home# ls
data  linux_practice
root@gourshab:/home# rmdir data
rmdir: failed to remove 'data': Directory not empty
root@gourshab:/home# rm -rf data
root@gourshab:/home# ls
linux_practice
root@gourshab:/home#
```

Assignment 2: File Editing & Text Processing

1. touch bio.txt
2. vi bio.txt
   Press **i** and write

   Name: Ravindra Gour
   Course: MCA
   City: indore

   Save file
   Presh - ESC
   :wq

3. Open file **vi bio.txt** then search **/Ravindra**

```
search hit BOTTOM, continuing at TOP
```

4. gg - move to start
   dG - delete all content
5. grep Ravindra bio.txt Search your name
   grep -v Ravindra bio.txt Lines **NOT containing name**

```
root@gourshab:/home# touch bio.txt
root@gourshab:/home# vi bio.txt
root@gourshab:/home# cat bio.txt
Name: Ravindra Gour
Course: MCA
City: Jamshedpur
root@gourshab:/home# vi bio.txt
root@gourshab:/home# vi bio.txt

[1]+  Stopped                 vi bio.txt
root@gourshab:/home# vi bio.txt
root@gourshab:/home# vi bio.txt
root@gourshab:/home# vi bio.txt
root@gourshab:/home# grep Ravindra bio.txt
Name: Ravindra Gour
root@gourshab:/home# grep -v Ravindra bio.txt
Course: MCA
City: Jamshedpur
root@gourshab:/home#
```

Assignment 3: Linux Permissions

1. touch script.sh
2. ls -l script.sh
3. chmod 755 script.sh
   chmod 764 script.sh
4. **Permission format**
   Owner | Group | Others
   Owner - 7 = rwx
   Group - 6 = rw
   Others - 4 = r
   r - Read only
   w - write only
   x - execute only
5. chmod +x script.sh
6. r=4, w=2, x=1?

Read (r = 4) - Allows a user to **view or read the file contents**.
Write (w = 2) - Allows a user to **modify or edit the file**.
Execute (x = 1) - Allows a file to **run as a program or script**.

7. Permission 777 - Give all the permission **Owner | Group | Others** to read , write and execute.
   Rwxrwxrwx =777

```
root@gourshab:/home# touch script.sh
root@gourshab:/home# ls -l script.sh
-rw-r--r-- 1 root root 0 Mar 15 16:06 script.sh
root@gourshab:/home# chmod 755 script.sh
root@gourshab:/home# ls -l script.sh
-rwxr-xr-x 1 root root 0 Mar 15 16:06 script.sh
root@gourshab:/home# chmod 764 script.sh
root@gourshab:/home# ls -l script.sh
-rwxrw-r-- 1 root root 0 Mar 15 16:06 script.sh
root@gourshab:/home# chmod +x script.sh
root@gourshab:/home# ls -l script.sh
-rwxrwxr-x 1 root root 0 Mar 15 16:06 script.sh
root@gourshab:/home#
```

Assignment 4 Linux Networking

Task : Check Network Configuration

1. ip a
2. **IP Address** -An IP Address (Internet Protocol Address) is a unique number given to every device connected to a network or the internet.

Types of IP Addresses

1. IPv4 -the older IP addressing system that uses 32-bit addresses written in four numbers separated by dots.
   Ex - 192.168.1.10
2. IPv6 - the newer IP addressing system that uses 128-bit addresses written in hexadecimal numbers separated by colons.
   Ex. 2001:db8::1

```
root@gourshab:/home# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1400 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:a8:b9:c2 brd ff:ff:ff:ff:ff:ff
    inet 172.18.155.31/20 brd 172.18.159.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fea8:b9c2/64 scope link
       valid_lft forever preferred_lft forever
```

Task : Test Network Reachability

3. ping google.com
4. T

   IP Address Resolved: 142.251.223.110

   Packet Loss: 0%

```
root@gourshab:/home# ping google.com
PING google.com (142.251.223.110) 56(84) bytes of data.
64 bytes from tzdela-ar-in-f14.1e100.net (142.251.223.110): icmp_seq=1 ttl=111 time=523 ms
64 bytes from tzdela-ar-in-f14.1e100.net (142.251.223.110): icmp_seq=2 ttl=111 time=229 ms
64 bytes from tzdela-ar-in-f14.1e100.net (142.251.223.110): icmp_seq=3 ttl=111 time=145 ms
^C
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 145.104/298.781/522.717/161.967 ms
root@gourshab:/home#
```

Task : Check Port Connectivity

5. telnet google.com 80

```
root@gourshab:/home# telnet google.com 80
Trying 142.250.182.206...
Connected to google.com.
Escape character is '^]'.

^]
HTTP/1.0 400 Bad Request
Content-Type: text/html; charset=UTF-8
Referrer-Policy: no-referrer
Content-Length: 1555
Date: Sun, 15 Mar 2026 11:09:19 GMT

<!DOCTYPE html>
<html lang=en>
  <meta charset=utf-8>
  <meta name=viewport content="initial-scale=1, minimum-scale=1, width=device-width">
  <title>Error 400 (Bad Request)!!1</title>
  <style>
    *{margin:0;padding:0}html,code{font:15px/22px arial,sans-serif}html{background:#fff;color:#222;padding:15px}body{margin:7% auto 0;max-width:390px;min-he
ight:180px;padding:30px 0 15px}* > body{background:url(//www.google.com/images/errors/robot.png) 100% 5px no-repeat;padding-right:205px}p{margin:11px 0 22px
;overflow:hidden}ins{color:#777;text-decoration:none}a img{border:0}@media screen and (max-width:772px){body{background:none;margin-top:0;max-width:none;pad
ding-right:0}}#logo{background:url(//www.google.com/images/branding/googlelogo/1x/googlelogo_color_150x54dp.png) no-repeat;margin-left:-5px}@media only scre
en and (min-resolution:192dpi){#logo{background:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.png) no-repeat 0% 0%/100% 100%;
-moz-border-image:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.png) 0}}@media only screen and (-webkit-min-device-pixel-rati
o:2){#logo{background:url(//www.google.com/images/branding/googlelogo/2x/googlelogo_color_150x54dp.png) no-repeat;-webkit-background-size:100% 100%}}#logo{d
isplay:inline-block;height:54px;width:150px}
  </style>
  <a href=//www.google.com/><span id=logo aria-label=Google></span></a>
  <p><b>400.</b> <ins>That's an error.</ins>
  <p>Your client has issued a malformed or illegal request.  <ins>That's all we know.</ins>
Connection closed by foreign host.
root@gourshab:/home#
```

6. Port **80** is HTTP web port.

7.

- Connection successful?

  The server is reachable and the port is open.
- Connection refused?

Server reachable but port closed.

- Connection timed out?
  Network or firewall blocking connection.

Task : Download Data Using curl

8. curl google.com -o google.html
9. ls , cat google.html -> check file

```
root@gourshab:/home# curl google.com -o google.html
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   219  100   219    0     0    336      0 --:--:-- --:--:-- --:--:--   336
root@gourshab:/home# ls
bio.txt  google.html  linux_practice  script.sh
root@gourshab:/home# cat google.html
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
root@gourshab:/home#
```

Assignment 5: Process & System Monitoring

1. Ps , ps aux

```
root@gourshab:/home# ps
    PID TTY          TIME CMD
    294 pts/0    00:00:00 bash
   1155 pts/0    00:00:00 vi
   1725 pts/0    00:00:00 ps
```

2. top

```
root@gourshab:/home# top
top - 17:05:39 up  2:07,  1 user,  load average: 0.03, 0.02, 0.00
Tasks:  26 total,   1 running,  24 sleeping,   1 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7790.2 total,   7171.4 free,    371.6 used,    247.2 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7265.4 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  165892  10696   8136 S   0.0   0.1   0:00.94 systemd
      2 root      20   0    3072   1920   1792 S   0.0   0.0   0:00.01 init-systemd(Ub
      6 root      20   0    3088   2040   1920 S   0.0   0.0   0:00.00 init
     60 root      19  -1   47820  14356  13588 S   0.0   0.2   0:00.38 systemd-journal
     86 root      20   0   22848   5504   4608 S   0.0   0.1   0:01.14 systemd-udevd
     89 systemd+  20   0   25672  13240   8960 S   0.0   0.2   0:00.17 systemd-resolve
```

3. df -h
4. free -h
5. Difference between ps and top
   ps - Static output , Shows snapshot.
   Top - Dynamic output , Shows real-time processes.
6. df -h shows disk space usage in human readable format (MB, GB).

```
root@gourshab:/home# df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G     0  3.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G   1% /mnt/wsl
drivers         476G  258G  219G  55% /usr/lib/wsl/drivers
/dev/sdd       1007G  1.6G  955G   1% /
none            3.9G  108K  3.9G   1% /mnt/wslg
none            3.9G     0  3.9G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.7M  3.8G   1% /init
none            3.9G  512K  3.9G   1% /run
none            3.9G     0  3.9G   0% /run/lock
none            3.9G     0  3.9G   0% /run/shm
none            3.9G   76K  3.9G   1% /mnt/wslg/versions.txt
none            3.9G   76K  3.9G   1% /mnt/wslg/doc
C:\             476G  258G  219G  55% /mnt/c
tmpfs           780M  8.0K  780M   1% /run/user/0
root@gourshab:/home# free -h
               total        used        free      shared  buff/cache   available
Mem:           7.6Gi       368Mi       7.0Gi       3.0Mi       247Mi       7.1Gi
Swap:          2.0Gi          0B       2.0Gi
```

Assignment 6: Shell Scripting (Basic)

Task 1: Simple Script

```
#!/bin/bash

echo "Welcome to Linux Training"
date
whoami
```

```
root@gourshab:/home# nano welcome.sh
root@gourshab:/home# chmod +x welcome.sh
root@gourshab:/home# ./welcome.sh
Welcome to Linux Training
Sun Mar 15 17:11:21 IST 2026
root
root@gourshab:/home#
```

Task 2: User Input Script

```
  GNU nano 6.2
#!/bin/bash

echo "Enter your name:"
read name

echo "Hello $name, welcome to Linux"
```

```
root@gourshab:/home# ls
bio.txt  google.html  linux_practice  script.sh  welcome.sh  welcome.sh.save
root@gourshab:/home# nano welcome.sh
root@gourshab:/home# ./welcome.sh
Enter your name:
Ravindra Gour
Hello Ravindra Gour, welcome to Linux
root@gourshab:/home#
```

Conditional & Loop Script

Task 3: Check number is less than 10 or not

```
#!/bin/bash

echo "Enter number:"
read num

if [ $num -lt 10 ]
then
echo "Number is less than 10"
else
echo "Number is greater than or equal to 10"
fi
```

```
root@gourshab:/home# ./greather.sh
Enter number:
8
Number is less than 10
root@gourshab:/home# ./greather.sh
Enter number:
78
Number is greater than or equal to 10
root@gourshab:/home#
```

Task 4: Loop Program

```
  GNU nano 6.2
#!/bin/bash

echo "Numbers from 1 to 10"

for i in {1..10}
do
echo $i
done

echo "Multiplication table of 5"

for i in {1..10}
do
echo "5 x $i = $((5*i))"
done
```

```
root@gourshab:/home# nano loop.sh
root@gourshab:/home# chmod +x loop.sh
root@gourshab:/home# ./loop.sh
Numbers from 1 to 10
1
2
3
4
5
6
7
8
9
10
Multiplication table of 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
root@gourshab:/home#
```

Assignment 7: Mini Project

```
#!/bin/bash

if [ -d "backup" ]
then
echo "Directory already exists"
else
mkdir backup
echo "Backup directory created"
fi
date
cp *.txt backup/
echo "Backup Completed"
```

```
root@gourshab:/home# nano backup.sh
root@gourshab:/home# chmod +x backup.sh
root@gourshab:/home# ./backup.sh
Backup directory created
Sun Mar 15 18:05:47 IST 2026
Backup Completed
root@gourshab:/home# ./backup.sh
Directory already exists
Sun Mar 15 18:06:27 IST 2026
Backup Completed
root@gourshab:/home#
```

Explanation: This shell script checks whether a backup directory exists.
If it does not exist, the script creates the directory. Then it prints the current date and time, copies all .txt files into the backup folder, and finally prints "Backup Completed".